

Two optimal Lorentz endpoint questions

Affirmative answers in arXiv:2010.05297

Run fa_banach_001 — agent_lane_03

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Status. `literature_already_answered`. This packet identifies and checks a later affirmative answer to both questions. It does not claim a new theorem.

The two source questions

Spector and Van Schaftingen, *Optimal embeddings into Lorentz spaces for some vector differential operators via Gagliardo's lemma*, arXiv:1811.02691, ask whether every homogeneous constant-coefficient elliptic canceling operator has the sharp Lorentz endpoint. Their Questions 1.1 and 1.2 ask, respectively, whether

$$\|u\|_{L^{n/(n-1),1}(\mathbb{R}^n)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n)} \quad (\text{ord } A = 1), \quad (\text{Q1})$$

$$\|D^{k-1}u\|_{L^{n/(n-1),1}(\mathbb{R}^n)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n)} \quad (\text{ord } A = k \geq 2). \quad (\text{Q2})$$

The dimensions in (Q1) are $n \geq 3$ and in (Q2) $n \geq 2$.

Question 1.1. *Let $n \geq 3$ and let V and E be finite-dimensional spaces. Further suppose that the first-order homogeneous linear differential operator with constant coefficients $A(D) : C_c^\infty(\mathbb{R}^n, V) \rightarrow C_c^\infty(\mathbb{R}^n, E)$ is elliptic and canceling. Can one show the existence of a constant $C > 0$ such that the inequality*

$$\|u\|_{L^{\frac{n}{n-1},1}(\mathbb{R}^n,V)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n,E)}$$

holds for every $u \in C_c^\infty(\mathbb{R}^n, V)$?

Another open question stems from the fact that higher-order operators satisfy Sobolev estimates [41, Theorem 1.3], Hardy inequalities [8, Theorem 1.2] and non-optimal Lorentz estimates [41, Theorem 8.5].

Question 1.2. *Let $n \geq 2$, $k \geq 2$ and let V and E be finite-dimensional spaces. Further suppose that the k -th order homogeneous linear differential operator with constant coefficients $A(D) : C_c^\infty(\mathbb{R}^n, V) \rightarrow C_c^\infty(\mathbb{R}^n, E)$ is elliptic and canceling. Can one show the existence of a constant $C > 0$ such that the inequality*

$$\|D^{k-1}u\|_{L^{\frac{n}{n-1},1}(\mathbb{R}^n,V)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n,E)}$$

holds for every $u \in C_c^\infty(\mathbb{R}^n, V)$?

The plan of the paper is as follows. In Section 2 we recall some requisite preliminaries regarding Lorentz spaces. In Section 3 we give a proof of Theorem 1 in the plane, as it illustrates well the idea of the more general Theorem 2. The main ingredient for higher dimensions is a version of an inequality of Loomis and Whitney [26] to nonorthogonal coordinate systems. We show how this can be obtained from a change of coordinates and a Lemma of Gagliardo in [17] in a presentation that tries to keep the geometric content of the inequality. In Section 5.1 we prove Theorem 2 from which Theorem 1 is deduced in Section 5.2. Finally, in Section 5.3 we prove a general result for elliptic and $(n-1)$ -canceling operators.

2. PRELIMINARIES

In the sequel, with an abuse of notation we utilize $\|\cdot\|$ to denote the norm in any

Exact later answer

Stolyarov, *Hardy–Littlewood–Sobolev inequality for $p = 1$* , arXiv:2010.05297v3 (2021), proves the required estimate in all orders. His abstract records as a particular case that, for every canceling elliptic differential operator A of order m ,

$$\|\nabla^{m-1}u\|_{L^{d/(d-1),1}(\mathbb{R}^d)} \lesssim \|Au\|_{L^1(\mathbb{R}^d)}. \quad (\text{S})$$

Taking $m = 1$ gives (Q1), since $\nabla^0 u = u$. Taking $m = k \geq 2$ gives (Q2), with equivalent finite-dimensional norms on the full $(k-1)$ st derivative tensor. Stolyarov’s statement holds for every $d \geq 2$; it therefore covers all dimensions asked in the source.

The theorem-to-question bridge

Here is the mechanism inside the later paper, included to rule out a merely notational match. For an order- m elliptic operator $A : V \rightarrow E$, Stolyarov's Example 1.4 assigns

$$\Omega(\zeta) = \text{Im } A(\zeta), \quad \zeta \in S^{d-1}.$$

The source cancellation hypothesis is exactly $\bigcap_{\zeta \in S^{d-1}} \Omega(\zeta) = \{0\}$. Hence Corollary 1.7 of the later paper, with $\alpha = 1$, gives

$$\|\mathbb{I}_1(Au)\|_{L^{d/(d-1),1}} \lesssim \|Au\|_{L^1}.$$

Ellipticity supplies a smooth homogeneous left inverse of the symbol: for $\xi \neq 0$, one may use

$$A(\xi)^\dagger = (A(\xi)^* A(\xi))^{-1} A(\xi)^*.$$

Consequently each component of $D^{m-1}u$ is a degree-zero smooth Fourier multiplier applied to $\mathbb{I}_1(Au)$. Such Calderón–Zygmund multipliers are bounded on $L^{d/(d-1),1}$, whose first exponent is strictly between 1 and ∞ . This proves (S), exactly with the source's domain, operator hypotheses, derivative order, and Lorentz index.

Proof intuition

An unconstrained L^1 datum can concentrate into a point mass, which is the endpoint obstruction for the Riesz potential. Cancellation forbids that common point-mass direction in the symbol images. Stolyarov proves that this is precisely enough to recover the strongest Lorentz target for $\mathbb{I}_1$. Ellipticity then reconstructs one fewer derivative of u from that potential using only order-zero singular integrals. Thus cancellation removes the L^1 concentration obstruction, while ellipticity turns the potential estimate into the desired differential estimate.

Scope and provenance

This is a complete affirmative resolution of both displayed source questions. The supporting paper was posted on 11 October 2020 and the bundled PDF is arXiv:2010.05297v3, dated 5 February 2021. Its bibliography includes the source paper, and its abstract explicitly states the full higher-order estimate. The bundled one-page source excerpt contains both questions; the four-page supporting excerpt contains the abstract, Example 1.4, the cancellation definition, and Corollary 1.7.

References. [1] D. Spector and J. Van Schaftingen, *Optimal embeddings into Lorentz spaces for some vector differential operators via Gagliardo's lemma*, arXiv:1811.02691, Questions 1.1–1.2. [2] D. Stolyarov, *Hardy–Littlewood–Sobolev inequality for $p = 1$* , arXiv:2010.05297v3; Sb. Math. 213 (2022), 844–889.